PROJECT REPORT

Inventory Manager Using Zoho Creator

A low-code stock management application built with Zoho Creator forms, reports, dashboards, workflows, and Deluge scripting

1. Project Description

The Inventory Manager is a low-code application developed using Zoho Creator. It helps organizations such as shops, school laboratories, and warehouses maintain product details, record stock received from suppliers, record stock issued, and monitor current stock levels.

The application uses Zoho Creator forms, reports, dashboards, workflows, and Deluge scripting to automate routine stock activities. It reduces manual registers and spreadsheets, keeps stock quantities accurate automatically, and helps the store team identify products that must be reordered before they run out.

2. Objectives

1. To maintain supplier and product information in a centralized database.
2. To record and manage stock received (Stock In).
3. To record and manage stock issued (Stock Out).
4. To calculate the current stock of every product automatically.
5. To prevent issuing more stock than is available.
6. To identify products that have reached their reorder level.
7. To generate reports and visual dashboards.
8. To automate notifications using workflows.
9. To explore AI-based demand analysis.

3. Forms Used

3.1 Suppliers Form

This form stores basic supplier information.

Field	Type	Settings
Supplier_Name	Single Line	Mandatory
Phone	Phone	Optional
Email	Email	Valid email format

3.2 Products Form

This form stores product details and the current stock level.

Field	Type	Settings
Product_Name	Single Line	Mandatory
SKU	Single Line	No duplicate values
Category	Dropdown	Electronics, Stationery, Furniture
Unit	Dropdown	Pcs, Box, Kg
Supplier	Lookup	Lookup to the Suppliers form
Reorder_Level	Number	Minimum stock before an alert is raised
Current_Stock	Number	Default 0; admin only, updated by workflow
Stock_Status	Single Line	In Stock, Low Stock, or Out of Stock (set by script)

3.3 Stock_In Form (Stock Received)

Field	Type	Settings
Product	Lookup	Lookup to the Products form
Quantity	Number	Mandatory; minimum 1
Received_Date	Date	Default value: today
Invoice_No	Single Line	Supplier invoice reference

3.4 Stock_Out Form (Stock Issued)

Field	Type	Settings
Product	Lookup	Lookup to the Products form
Quantity	Number	Mandatory; minimum 1
Issued_Date	Date	Default value: today
Issued_To	Single Line	Person or department receiving the stock

4. Dataset Used

A sample dataset containing eight products is used to test the application. Current stock is the total received minus the total issued.

SKU	Product	Category	Total In	Total Out	Current Stock	Reorder Level
STN-001	Notebook	Stationery	100	60	40	10
ELC-001	Laptop	Electronics	30	6	24	5
ELC-002	Monitor	Electronics	25	7	18	5
ELC-003	Mouse	Electronics	40	25	15	5
STN-002	Stapler	Stationery	20	6	14	5
FUR-001	Chair	Furniture	15	3	12	3
ELC-004	Keyboard	Electronics	30	26	4	5
STN-003	Pen Box	Stationery	20	17	3	5

Note: These are sample values for demonstration. Current stock should always be calculated from the actual Stock_In and Stock_Out records. The sample data contains 8 products, 130 units in total, and 2 products at or below their reorder level.

5. Reports Generated

5.1 Products Report

Displays product names, SKUs, categories, suppliers, current stock, and reorder levels.

5.2 Stock In History

Displays every stock receipt grouped by product, with quantity, date, and invoice number.

5.3 Stock Out History

Displays stock issued, grouped by month and by the person or department it was issued to.

5.4 Low Stock Report

Identifies products whose current stock is less than or equal to the reorder level (Current_Stock <= Reorder_Level).

5.5 Stock by Category Report

Summarizes the total units in stock for each category.

5.6 Dashboard

Displays key figures and charts in a single interface:

- KPI cards for total products, units in stock, low-stock items, and units issued this month.
- A column chart of current stock by product.
- A pie chart of stock by category.
- A line chart of stock-out quantity by month.

6. Three Deluge Scripts

The following examples use Zoho Creator's Deluge scripting language. Form and field link names must match the actual application configuration.

Script 1: Recalculate Current Stock (custom function recalcStock)

Purpose: Calculate the current stock of a product as total received minus total issued, store it in the Products form, and send an alert when the product reaches its reorder level.

```
void recalcStock(int productId)
{
    inQty  = ifnull(Stock_In[Product == productId].sum(Quantity), 0);
    outQty = ifnull(Stock_Out[Product == productId].sum(Quantity), 0);

    prod = Products[ID == productId];
    prod.Current_Stock = inQty - outQty;

    if(prod.Current_Stock <= prod.Reorder_Level)
    {
        sendmail
        [
            from: zoho.adminuserid
            to: zoho.adminuserid
            subject: "Low stock: " + prod.Product_Name
            message: prod.Product_Name + " is at " + prod.Current_Stock + " units."
        ]
    }
}
```

Implementation note: Create this function under Workflow > Functions. Call it with `thisapp.recalcStock(input.Product);` from the On Success event of Add, Edit, and Delete on both Stock_In and Stock_Out. Because stock is always rebuilt from the full history, it stays correct even when records are edited or deleted.

Script 2: Prevent Issuing More Than the Available Stock

Purpose: Block a Stock_Out entry when the quantity is greater than the stock on hand.

```
prod = Products[ID == input.Product];
if(input.Quantity > prod.Current_Stock)
{
    alert "Only " + prod.Current_Stock + " units available in stock.";
    cancel submit;
}
```

Implementation note: Configure this in the On Validate event of Add on the Stock_Out form. For edits, add the old quantity back to the available stock before comparing, and disable the Product field on edit.

Script 3: Classify the Stock Status

Purpose: Mark each product as In Stock, Low Stock, or Out of Stock.

```
if(input.Current_Stock <= 0)
{
    input.Stock_Status = "Out of Stock";
}
else if(input.Current_Stock <= input.Reorder_Level)
{
    input.Stock_Status = "Low Stock";
}
else
{
    input.Stock_Status = "In Stock";
}
```

Implementation note: Create the Stock_Status field in the Products form. Execute this logic when the product is saved, and after recalcStock updates Current_Stock, using field link names that match your application.

7. Workflow Automation

The application uses automated workflows to reduce manual work.

Workflow 1: Stock Update

- **Trigger:** A Stock_In or Stock_Out record is created, edited, or deleted.
- **Action:** Call recalcStock for the product.
- **Result:** Current_Stock is always accurate without manual calculation.

Workflow 2: Over-Issue Prevention

- **Trigger:** A Stock_Out record is submitted (On Validate).
- **Condition:** Quantity is greater than the available stock.
- **Action:** Show an alert and cancel the submission.
- **Result:** Stock can never go negative.

Workflow 3: Low Stock Alert

- **Trigger:** Current_Stock is updated.
- **Condition:** Current_Stock is less than or equal to Reorder_Level.
- **Action:** Update the stock status and send an email to the administrator.
- **Result:** The store team can reorder before the product runs out.

Workflow 4: Daily Low Stock Digest (Schedule)

- **Trigger:** Runs once every day.
- **Action:** Collect all products at or below the reorder level and email one summary.
- **Result:** One daily summary instead of many separate emails.

8. AI Features

The initial version uses rule-based automation. The following AI features can be added as optional enhancements.

8.1 Demand Forecasting

An external machine learning model can analyze past stock-out records to estimate how much of each product will be needed in the coming weeks.

8.2 Smart Reorder Suggestions

AI can suggest reorder levels and order quantities based on consumption patterns and supplier delivery times.

8.3 Unusual Activity Detection

AI can flag unusual patterns, such as a sudden large stock-out or repeated adjustments, for manual review.

8.4 AI Integration

An external AI service can be integrated using an API and Zoho Creator connections or supported Deluge integration methods (for example, the invokeUrl function).

Note: These AI capabilities require an actual model or AI service integration. They are not automatically available just because the application is built in Zoho Creator.

9. Application Workflow

1. The administrator adds suppliers and products, setting a reorder level for each product.
2. Details are stored in the application.
3. The storekeeper records stock received in the Stock_In form.
4. The recalcStock function updates the current stock of the product.
5. The storekeeper records stock issued in the Stock_Out form.
6. Validation blocks the entry if the quantity exceeds the available stock.
7. Current stock is recalculated and the stock status is updated.
8. Workflows send alerts when a product reaches its reorder level.
9. Reports display stock levels and stock movement.
10. The dashboard presents summaries and charts for monitoring.

10. Implementation Steps

1. Open the Zoho Creator website and sign in.
2. Create a new application named Inventory Manager.
3. Create the Suppliers form.
4. Create the Products form with a Supplier lookup, Reorder_Level, and an admin-only Current_Stock field.
5. Create the Stock_In form with a Product lookup field.

6. Create the Stock_Out form with a Product lookup field.
7. Add sample suppliers, products, and stock records.
8. Create the product, stock history, and low stock reports.
9. Create the dashboard page and add KPI cards and charts.
10. Create the recalcStock function and the validation and status scripts.
11. Create workflows for stock update, over-issue prevention, and low stock alerts.
12. Test the forms, reports, calculations, and notifications, including edits and deletes.
13. Configure user permissions and publish or share the application as required.

11. Expected Output

The completed application is expected to provide:

- A centralized database of suppliers and products.
- A record of every stock receipt and stock issue.
- Automatic current stock calculation for every product.
- Prevention of issuing more stock than is available.
- Identification of products that need reordering.
- Reports and dashboard charts for the store manager.
- Automated alerts through configured workflows.

For the sample dataset, Keyboard (4 units, reorder level 5) and Pen Box (3 units, reorder level 5) are at or below their reorder levels, so they will appear in the Low Stock report and trigger alerts. Notebook has 40 units against a reorder level of 10, so it is classified as In Stock. The dashboard is expected to show 8 products and 130 units in total.

12. Future Enhancements

1. Integrate an AI-based demand forecasting model.
2. Add barcode or QR code scanning for fast stock entry and issue.
3. Add a purchase-order module with an approval flow.
4. Support multiple warehouses and stock transfers.
5. Add role-based access for administrators, store managers, storekeepers, and viewers.
6. Post low-stock alerts to Zoho Cliq or Slack through Zoho Flow.
7. Generate downloadable stock and purchase reports.
8. Integrate with accounting or ERP systems such as Zoho Books through APIs.

13. Conclusion

The Inventory Manager demonstrates how Zoho Creator can be used to develop a stock management application with minimal coding. The application centralizes supplier and product information, stock receipts, and stock issues while providing reports, dashboards, and automated workflows.

The use of Deluge scripting enables accurate stock calculation, validation, and rule-based alerts. Calculating stock from the full transaction history keeps the numbers correct even when records are edited or deleted. With additional AI integration, the system can be extended to provide demand forecasting and smarter reorder suggestions. Overall, the project improves record keeping, reduces repetitive manual work, and supports data-driven inventory decisions.